

Cardiovascular Disease Prediction Using Statistical Learning and Neural Network Models

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Abstract

This study predicts cardiovascular disease from a tabular clinical dataset with $n = 70,000$ patient records [3]. We cleaned implausible values, converted age to years, computed body mass index, and used a stratified train-test split. After cleaning, 69,546 records remained. We compared five supervised methods: Ridge logistic regression, Lasso logistic regression, k -nearest neighbors, random forest, and a multilayer perceptron. Stratified 5-fold cross-validation selected the hyperparameters on the training set, with AUC as the main score [12]. Scaling was refitted inside each fold for the models that need it. A separate test set measured final performance and was not used for model choice. The main predictors were systolic and diastolic blood pressure, age, BMI, and cholesterol. Glucose and physical activity added a smaller signal. Random forest gave the highest test AUC, 0.804, only slightly above the MLP and the simpler models. Non-linear methods gave a small predictive gain, not a large change. Penalized logistic regression remained easier to explain through its coefficients. For medical use, model choice must also include calibration, transparency, the decision threshold, and the costs of false negatives and false positives.

Keywords: cardiovascular risk prediction; statistical learning; logistic regression; random forest; neural network; model interpretability; medical decision support.

1 Introduction

Cardiovascular disease is a major public health problem and the leading cause of death worldwide [2]. Predictive tools can help identify people at high risk and support screening decisions. We define a binary classification problem with $Y = \text{cardio} \in \{0, 1\}$ and estimate $\hat{p}(X) = P(Y = 1 | X)$. The predictors X include age, height, weight, gender, systolic and diastolic blood pressure, cholesterol, glucose, smoking, alcohol use, and physical activity. Each record gives one simplified clinical snapshot of a patient.

The study has two goals. The first is to build a complete and reproducible analysis from the raw Kaggle data. This includes converting numeric columns, correcting implausible blood-pressure values, converting age to years, creating BMI, removing invalid records, splitting the data, scaling inside the validation pipeline, selecting hyperparameters, and evaluating the final models [5, 9]. The second goal is to compare simple interpretable methods with more flexible ones. Ridge and Lasso give coefficient-based logistic models, KNN uses local neighborhoods, random forest uses many decision trees, and the MLP gives a neural-network benchmark. We study predic-

tive performance, stability, interpretability, and clinical relevance together.

2 Dataset and Exploratory Analysis

2.1 Dataset structure

The original Kaggle dataset contains $n = 70,000$ observations and no explicit missing values [3]. Each row represents one patient record, and the dataset has no longitudinal follow-up information.

The variables are age in days, height, weight, gender, ap_hi, ap_lo, cholesterol, gluc, smoke, alco, active, and the binary target cardio. When present, id is only an identifier and is removed from X . The predictors are quantitative, binary, or ordinal. Blood pressure, height, and weight contained negative values, impossible values, reversed pressures, and likely unit errors. These issues are hidden by the absence of explicit missing values, so direct model fitting would still use invalid measurements. After correction and filtering, $n = 69,546$ observations remained [1]; 454 records, or 0.65%, were removed. We converted age to years, computed BMI, and removed raw weight from the final feature set to limit redundancy with BMI while keeping height.

2.2 Main variables associated with cardiovascular risk

The two target groups show clear differences in their empirical distributions. On average, patients with $\text{cardio}=1$ are older and have higher systolic pressure, diastolic pressure, and BMI. Abnormal cholesterol and glucose levels are also more common in this group. The boxplots and density curves shift in the same direction, so the differences are not caused by a few isolated values. These patterns match established cardiovascular risk factors [2].

After cleaning, the main Pearson correlations are $\rho(Y, \text{ap_hi}) \simeq 0.433$, $\rho(Y, \text{ap_lo}) \simeq 0.347$, $\rho(Y, \text{age}) \simeq 0.238$, $\rho(Y, \text{cholesterol}) \simeq 0.221$, and $\rho(Y, \text{BMI}) \simeq 0.195$. These are marginal linear associations with the binary target. They support prediction, but no causal effect is estimated.

Figure 1 shows the cleaned correlation structure. Blood pressure, age, cholesterol, and BMI have the strongest links with the target. Density plots and boxplots also shift toward higher systolic pressure, diastolic pressure, and age for patients with cardiovascular disease. Glucose has a smaller positive association. Gender, smoking, alcohol use, and physical activity add less marginal information. Their coding is coarse and gives no duration, intensity, dose, or medical history, so weak measured associations

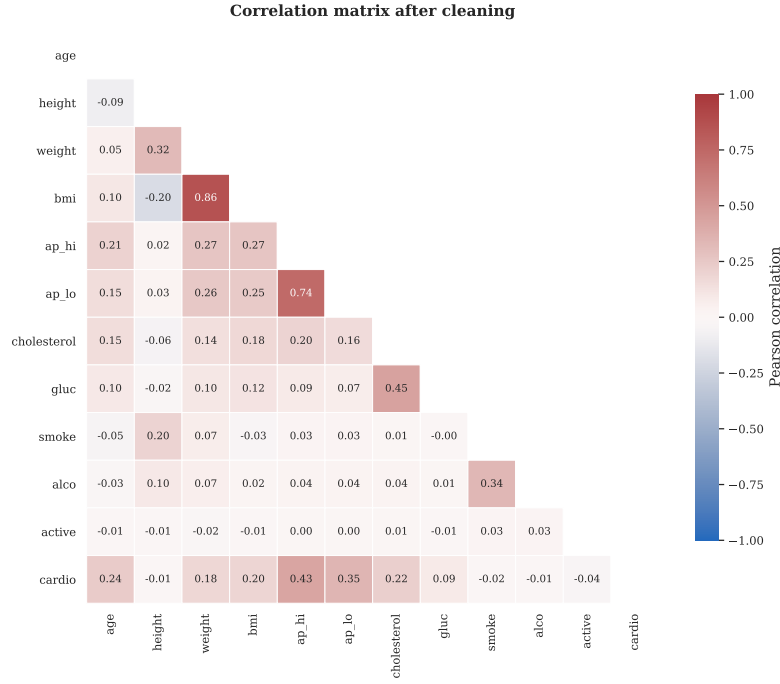


Figure 1: Correlation matrix after cleaning and feature creation. Systolic pressure, diastolic pressure, age, cholesterol, and BMI have the largest simple correlations with the target.

may partly reflect limited data quality.

2.3 Outliers and atypical distributions

The raw `ap_hi` and `ap_lo` distributions contained negative values, zeros, very small values, and extreme values outside a plausible clinical range. We treated them as likely entry or unit errors, not as ordinary medical outliers. A negative value was made positive only when the sign error was clear. A factor-of-ten error was corrected only when the new value was plausible. If `ap_lo > ap_hi`, the values were swapped only when the new pair was clinically plausible. Ambiguous cases were removed instead of being changed by a broad automatic rule.

After cleaning, $60 \leq \text{ap_hi} \leq 240$ mmHg within the allowed range $[60, 250]$, with mean 127.04 mmHg. Also, $40 \leq \text{ap_lo} \leq 150$ mmHg, with mean 81.57 mmHg. Mean age was 53.30 years. BMI had mean 27.42 and remained right-skewed, which is common for a positive body-size measure.

2.4 Correlations between explanatory variables

`ap_hi` and `ap_lo` are positively correlated because both measure blood pressure. Weight and BMI are strongly related, so including both can repeat much of the same information. Cholesterol and glucose have a moderate association, which is consistent with shared metabolic profiles. The full predictor set shows no severe multicollinearity outside these expected pairs. We removed raw weight after computing BMI because BMI combines weight and height in a more useful clinical measure:

$$\text{BMI} = \frac{\text{weight}}{(\text{height}/100)^2}.$$

This choice was checked in the notebook [1]. Weight alone, BMI alone, and weight with height and BMI gave almost identical AUC values across the tested models. Keeping BMI and height and dropping raw weight preserved prediction, reduced redundant input, and made the final feature set easier to explain.

2.5 Target balance

The target is almost balanced: $n_0 = 34,820$, $n_1 = 34,726$, $P(Y = 0) = 0.5007$, and $P(Y = 1) = 0.4993$. A classifier that always predicts the majority class has accuracy 0.5007, so class balance alone cannot produce a high score. Stratification can preserve these proportions in training and test data.

Accuracy alone hides the type of error. We also report recall = $\text{TP}/(\text{TP} + \text{FN})$, precision = $\text{TP}/(\text{TP} + \text{FP})$, F1-score, AUC, and the confusion matrix. A false negative is a diseased patient predicted as disease-free, and it may cost more than a false positive. A false positive can still create unnecessary tests or referrals. The decision threshold t must reflect these clinical costs; $t = 0.5$ is only the default.

3 Preprocessing Pipeline

3.1 Cleaning and feature creation

Before model fitting, a reproducible cleaning pipeline converted columns imported as objects, including `weight`, `ap_hi`, and `ap_lo`, to numeric values. The same ordered steps were then applied to every record, which makes the cleaning rules explicit and repeatable.

We corrected negative pressures only when the sign error was clear, and factor-of-ten errors only when the cor-

rected value was clinically plausible. If $ap_lo > ap_hi$, the pair was swapped only when the result made sense. We kept records satisfying $60 \leq ap_hi \leq 250$, $40 \leq ap_lo \leq 150$, $130 \leq height \leq 210$ cm, $40 \leq weight \leq 200$ kg, and $15 \leq BMI \leq 50$. These bounds remove measurements that are not plausible for the intended adult clinical dataset. Weight stayed continuous; binning would remove its order, distances, and physical scale and would add arbitrary cut points. In total, 454 records, or 0.65%, were removed.

3.2 Train-test split

The cleaned data used a stratified 80%/20% split on cardio: $n_{train} = 55,636$ and $n_{test} = 13,910$. Stratification keeps nearly the same class balance in both subsets. The positive-class proportions were 0.4993 in training and 0.4994 in testing, so any score difference between the two sets cannot be explained by a large change in prevalence.

The test set was used only for the final evaluation [5,9]. Hyperparameters, scaling, and candidate thresholds were chosen from the training data with stratified 5-fold cross-validation. The test labels were not used during these choices. A fixed `random_state` made the split, cross-validation, and stochastic models reproducible.

3.3 Standardization

The continuous variables age, height, BMI, `ap_hi`, and `ap_lo` were standardized by

$$X_{std} = \frac{X - \mu_{train}}{\sigma_{train}},$$

where μ_{train} and σ_{train} came from the current training fold only. The scaler was refitted in every fold, so validation data could not affect the transformation. For the final evaluation, one scaler was fitted on the full training set and applied unchanged to the test set.

Scaling is important for KNN, penalized logistic regression, and the MLP because distances, penalties, and gradient updates depend on feature size. Binary variables kept their 0/1 coding. The ordered levels of `cholesterol` and `gluc` were kept as ordinal values. One-hot encoding was also tested and gave almost the same AUC values, so we used the simpler ordinal form with fewer columns.

3.4 Avoiding information leakage

Every data-dependent preprocessing step was fitted only on training observations. Each validation fold used parameters from its training folds, and the final test set was transformed without refitting. This blocks information leakage: validation and test values cannot influence the transformations used to predict them. The procedure gives a more realistic estimate for unseen patients [5,9].

4 Models

We compared Ridge logistic regression, Lasso logistic regression, KNN, random forest, and an MLP. Ridge and Lasso are interpretable linear baselines; KNN is a local non-parametric model; random forest and MLP

add nonlinear flexibility [5,9]. These five methods cover coefficient-based, neighbor-based, tree-based, and neural-network predictions. Stratified 5-fold cross-validation selected the hyperparameters on the training set. Scaling stayed inside the pipeline for scale-sensitive models and was refitted in each fold. AUC was the main selection score because it measures ranking over all decision thresholds and does not depend on choosing $t = 0.5$ [12].

4.1 Ridge logistic regression

Ridge logistic regression adds an L_2 penalty to the negative log-likelihood and stabilizes coefficients when predictors are correlated [6]:

$$\min_{\beta} \left[-\ell(\beta) + \lambda \sum_j \beta_j^2 \right].$$

We tested $C = 1/\lambda$ over

$$C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}.$$

Small C means strong regularization and large C means weak regularization. Cross-validation selected $C = 10$, with mean AUC 0.7927 [1].

4.2 Lasso logistic regression

Lasso logistic regression uses an L_1 penalty. It can set some coefficients exactly to zero and create a sparse model [7]:

$$\min_{\beta} \left[-\ell(\beta) + \lambda \sum_j |\beta_j| \right].$$

Cross-validation selected $C = 100$, with mean AUC $\simeq 0.7927$ [1]. Since $C = 1/\lambda$ is large, the penalty is weak. The fitted model stayed close to Ridge, most predictors remained active, and the final solution had little sparsity.

4.3 k -nearest neighbors

KNN classifies a point from its local neighborhood in feature space [8]. With uniform weights, $\hat{p}(x) = k^{-1} \sum_{i \in N_k(x)} y_i$, and the class is obtained by comparing this probability with a threshold. After standardization, we tested several values of k and both weighting rules. The best model used $k = 100$ with uniform weights and reached mean cross-validation AUC 0.7962. Smaller k uses fewer patients, gives a more irregular boundary, and has higher variance. Larger k gives a smoother boundary, but a very large neighborhood can hide local structure and underfit.

4.4 Random forest

A random forest averages many randomized decision trees, which lowers the variance of one tree and keeps nonlinear splits [4]. Each tree can use different observations and predictor choices, and the final probability averages their outputs. The grid covered the number of trees, maximum depth, and minimum samples per leaf. The selected model used 300 trees, depth 12, and `min_samples_leaf=10`, with mean cross-validation AUC

0.8029. It can model thresholds and interactions, such as age with blood pressure, and remains easier to inspect than a neural network.

4.5 Multilayer perceptron

The MLP uses fully connected layers, ReLU activations, and a sigmoid output $\hat{p}(Y = 1 | X)$. Backpropagation fits this flexible nonlinear model [10, 11]. We tested one-layer architectures (50) and (100) and the two-layer architecture (50, 25). The best network had two hidden layers, (50, 25), with $\alpha=0.001$ and learning rate 10^{-3} . Its mean cross-validation AUC was 0.8016 [1], close to the random-forest value.

The model minimizes binary cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{p}_\theta(x_i) + (1 - y_i) \log(1 - \hat{p}_\theta(x_i))].$$

Early stopping monitored validation performance and ended training when it stopped improving. This limits unnecessary updates and reduces overfitting to the training sample.

5 Results

We compared the selected models once on the untouched test set. At threshold $t = 0.5$, classes were $\hat{y} = 1\{\hat{p} \geq 0.5\}$. Accuracy, precision, recall, and F1-score therefore depend on this threshold. AUC was reported separately because it measures ranking across thresholds [12]. Recall is especially important in medicine because a false negative is a diseased patient who is not detected and may not receive further assessment.

Table 1: Test-set performance of the selected models. Cross-validation AUC is the mean $\pm$ standard deviation over 5 folds.

Model	Acc.	Prec.	Recall	F1	Test AUC	CV AUC
Random forest	0.738	0.761	0.692	0.725	0.804	0.803 ± 0.002
MLP (50, 25)	0.738	0.771	0.675	0.720	0.803	0.802 ± 0.003
KNN ($k = 100$)	0.737	0.769	0.678	0.720	0.798	0.796 ± 0.002
Ridge	0.732	0.763	0.672	0.715	0.796	0.793 ± 0.003
Lasso	0.732	0.763	0.672	0.715	0.796	0.793 ± 0.003

Table 1 shows a small advantage for random forest and MLP over the linear models, with KNN between them. Test AUC ranges only from 0.796 to 0.804, so the absolute differences are modest. Cross-validation and test AUC are also close for every method. The protocol therefore appears stable, and no model shows clear overfitting on this split.

5.1 Calibration of predicted probabilities

Calibration asks whether predicted probabilities match observed frequencies [14]. We used the Brier score, $BS = n^{-1} \sum_{i=1}^n (y_i - \hat{p}_i)^2$, where smaller values are better. Random forest gave 0.1792 and Ridge logistic regression gave 0.1848. Random forest had the best ranking and a slightly lower Brier score, but the difference was small. A full analysis would also examine calibration curves

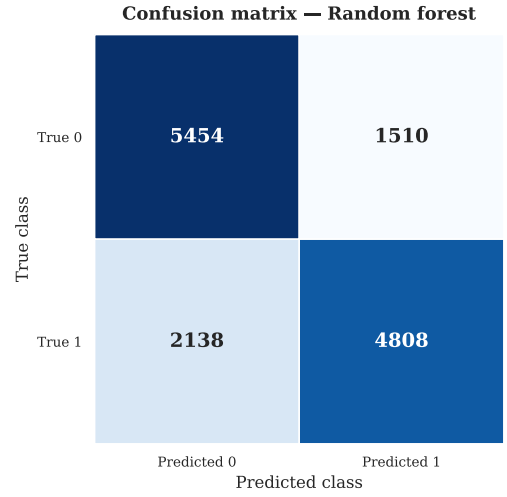


Figure 2: Random-forest confusion matrix on the test set at $t = 0.5$. False negatives are diseased patients predicted as disease-free.

and possibly recalibrate the model. More work is needed before using these probabilities as individual medical risks.

5.2 Best predictive model

Random forest has the highest test AUC. Its test scores are accuracy = 0.738, precision = 0.761, recall = 0.692, F1-score = 0.725, and AUC = 0.804. At the default threshold $t = 0.5$, the confusion matrix contains the numbers of true negatives, false positives, false negatives, and true positives:

$$\begin{pmatrix} 5454 & 1510 \\ 2138 & 4808 \end{pmatrix},$$

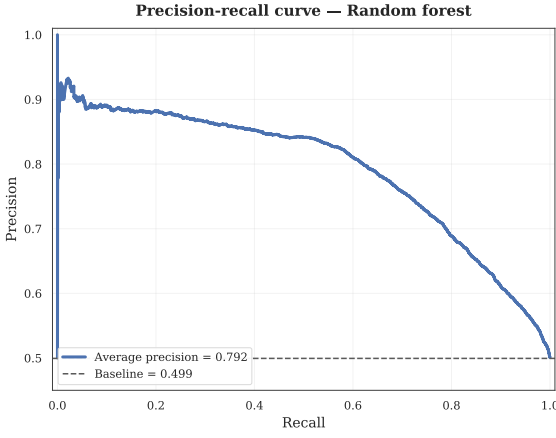
where rows are true classes and columns are predicted classes. Thus 5454 patients are true negatives, 1510 are false positives, 2138 are false negatives, and 4808 are true positives. The recall is $4808 / (4808 + 2138) \simeq 0.692$.

Figure 2 shows the errors at $t = 0.5$. If the priority is to reduce false negatives, the threshold can be lowered so that more patients are classified as positive. At $t = 0.40$, recall rises to 0.781, precision falls to 0.708, and false positives increase. This gives better sensitivity but more unnecessary referrals. The chosen t must reflect the relative costs C_{FN} and C_{FP} in the intended medical setting.

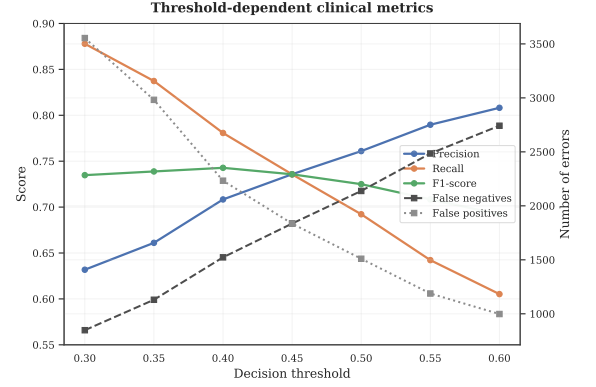
When false negatives have a high cost, the precision-recall and threshold curves in Figure 3 are more useful than accuracy alone [13]. They show precision, recall, F1-score, false negatives, and false positives as functions of t . This makes the clinical effect of changing the threshold visible before deployment.

5.3 Stability of the results

Cross-validation and test AUC are close. The large sample, with more than 55,000 training observations, and the nearly balanced target also reduce sampling variation. Ridge and Lasso are stable because they solve convex regularized problems. Random forest stabilizes as many



(a) Precision-recall curve.



(b) Threshold-dependent metrics.

Figure 3: Clinical threshold analysis for the selected random forest. Lowering t raises recall and also raises the number of false positives.

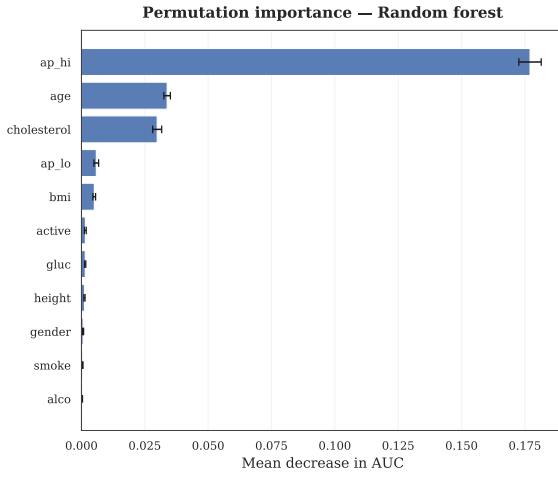


Figure 4: Permutation importance for the selected random forest. Systolic pressure, age, cholesterol, diastolic pressure, and BMI cause the largest mean decrease in AUC when shuffled.

trees are averaged, although its result still depends on the random seed and tuning choices. MLP results can vary more with initialization, mini-batch order, and optimization.

5.4 Interpretability

The best predictive model is not always the easiest to explain. Ridge and Lasso give standardized coefficients with clear signs and one global formula. The AUC-optimal Lasso penalty is weak, so its solution is close to Ridge and gives little variable selection. Figure 4 shows random-forest permutation importance. For feature j , it measures $\Delta\text{AUC}_j = \text{AUC} - \text{AUC}_{\pi(j)}$ after shuffling that feature while leaving the others unchanged. Systolic pressure, age, cholesterol, diastolic pressure, and BMI have the largest values. This measure is more informative here than impurity importance because it directly records the loss in predictive performance on held-out data [4, 15].

Random forest is a middle option: global importance scores give useful information about the model as a whole, but individual predictions are less transparent than logistic regression. The MLP is the hardest to explain because its internal weights and hidden units have no simple clinical meaning. In medicine, a small loss in accuracy can be acceptable when a model must be explained to patients, justified to clinicians, or checked in an audit.

6 Discussion

All methods identify the same main predictors: age, systolic and diastolic pressure, BMI, cholesterol, and, with smaller effects, glucose and physical activity. This agreement across correlations, linear coefficients, and permutation importance strengthens the descriptive result. It also matches known cardiovascular risk factors [2]. The behavioral variables add less information because binary coding misses duration, intensity, dose, and past exposure.

The study estimates associations for prediction. Since the data are observational, the models show which variables help classify patients in this sample. They do not identify the effect of changing one variable on a patient's future risk, and they cannot separate direct effects from shared causes. A noisy or coarsely measured predictor can receive low importance even when it matters clinically. Redundancy can also split importance across several correlated variables.

6.1 Comparison of methods

For maximum interpretability, Ridge and Lasso are the main choices. They are fast, stable, convex, and their standardized coefficients are easy to communicate. The sign of each coefficient gives the direction of the model association, and its size can be compared across standardized continuous variables. KNN is easy to describe, but its predictions depend strongly on the distance, k , the scaling, and the geometry of the feature space.

Random forest gives the best prediction in this study. It captures thresholds, interactions, and nonlinear effects with a manageable computational cost for 70,000 rows. Global permutation importance provides some interpretation, even though individual paths through hundreds of trees remain complex. The MLP has similar flexibility, but it is more expensive, less transparent, and more sensitive to hyperparameters. On this structured tabular dataset, the MLP serves mainly as a nonlinear benchmark.

6.2 Relevance of neural networks

The neural network was included as a nonlinear benchmark. The dataset has only a modest number of structured clinical variables, so a large gain over trees or linear models was unlikely. Neural networks are often most useful when the input contains many complex features, while this table contains a small set of hand-recorded variables. Its AUC was close to the random forest AUC but not higher. Because it is harder to explain and tune, random forest or penalized logistic regression fits this task better.

6.3 Dataset limitations

The dataset gives only one simplified snapshot per patient. It lacks medication, family history, disease severity, diagnosis date, previous cardiovascular events, and smoking duration. Smoking, alcohol use, and physical

activity are also coded in a coarse binary form. These gaps limit prediction, interpretation, and any attempt to transfer the model to a richer clinical setting. Any clinical use would require external validation on another population and a detailed calibration study.

7 Conclusion

The clinical and behavioral variables contain a real but incomplete signal for cardiovascular disease. Blood pressure, age, cholesterol, and BMI are the most informative predictors across the exploratory analysis and fitted models. Penalized logistic regression already performs well, so simple relations capture much of the available signal. Random forest has the highest test AUC, but the gain over the other models is small: 0.804 versus 0.796–0.803. It is a good choice when prediction is the main goal because it captures nonlinear effects and interactions and is easier to inspect than the MLP. Ridge and Lasso remain strong references when clinical explanations, coefficient signs, or auditability matter. The threshold t should not be fixed automatically at 0.5: a lower value can be better when false negatives cost more than false positives, but it also creates more false alarms. Before practical use, every model needs external validation, threshold selection for the target setting, and a stronger calibration study. Overall, random forest is the strongest predictor here, and penalized logistic regression is the simpler, more transparent benchmark.

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